

DT

2 what is mongoDB

-> a popular NOSQL database that stores data in a JSON like format

mongodb is a popular open source NoSQL document db program

it uses json like documents with optional schemas

mongodb is a cross platform,document oriented db that provide high performance

10 explain features of mongoDB

-> flexible schema - stores data in flexible,like json bson format

scalable

high performance - supports data models
easy data retrieval

high availability - provides replication
sharding - splitting a single database

into smaller

5 what is CRUD operation

-> create,read,update,delete operations that can be performed on data in a database

12 list some of the popular graph database

-> graph db is a database that store data as nodes and edges such as Neo4j and amazon

Neptune

neo4j - a highly popular native graph db

amazon Neptune - a fully managed graph db

arangodb - a multimodel db that supports

graph,document,key-value data

orientDB - a multimodel db with strong graph capability

15 explain peer to peer replication

-> method of data replication where data is replicated between nodes

in a network without a central server

also known as multi master replication

this differ from single- master

replication

data changes can originate at any peer

17 NoSQL databse are schemaless comment

-> NoSQL database do not require a predefined schema allowing for flexible data modelling

no need to predefine the schema in the db before inserting the data

different documents within the same collection can have different fields and structure

11 explain the features of bootstrap

-> responsive design /grid system - provide mobile flexbox grid for building layouts

pre-built UI components - ready to use components like buttons ,navigation bars,cards,forms

css utility classes- a vast collection of utility classes for applying styles like spacing,color

extensive documentation - well organized

documentation

13 list some benefits of using bootstrap

-> rapid development - provides pre-written code, components, front-end development

responsive design - built with a mobile first approach

consistency - standardized component

browser compatibility - handles many cross browser

6 what is a bootstrap container

-> a container is a basic building block of the bootstrap grid system

it is a fundamental layout component

it serves as the most basic layout element

it is required when using the default grid system

containers two main types:

.container - sets a responsive, fixed width

.container-fluid - always 100%

3 explain any two ways to include bootstrap in your project

-> bootstrap can be included in a project using a CDN or by downloading the files locally

using a content delivery network(CDN) -

link directly to bootstrap compiled css and js files hosted online

this requires no download and is typically the quickest way to get started

downloading the files -

download the compiled css and js files
from the official bootstrap website and
host them locally within your project
directory

16 explain bootstrap grid system

-> a grid system that allows for responsive
design and layout

is a layout utility built with flexbox
it uses a containers, rows, cols, align
the core principle is row is divided into
12 cols which can be used individually
provides a responsive way to layout
content using rows and cols

4 how to include CSS in the webpage

-> css can be included in a webpage using the
3 ways

external stylesheets - linking a separate
.css file using the <link> tag within
the <head> section
of the html document

eg. <link
rel="stylesheet" href="styles.css">

internal css - placing css rules within a
<style> tag in the <head>
section of the html

document

inline css - applying styles directly to
individual html elements using the

style attribute

eg. <p style="color:blue;">

14 explain basic components of a CSS3 transition

-> a transition is a way to control the speed and timing of CSS property changes

transition-property: specifies the css property the transition effect

eg. color, opacity, transform

transition-duration: defines how many seconds the transition takes to complete

transition-timing : specifies the speed curve of the transition eg.

ease, ease-in, linear

transition-delay : defines when the transition will start

18 how do you make images responsive

-> images that can be resized and adapted to different screen sizes and devices

using css max-width : the most common technique is setting img

{max-width:100%;height:auto;}.

using html srcset attribute : this attribute allows multiple images files for different screens

19 create an emp collection with mentioned fields

emp(eno,ename,salary,desig,dept,{deptno,dep

```

tname,location},roject:{pname,hrs})
    I)display all employee with
salary>50000 and salary<80000
    II)Update no.of hrs to 7 forpname
-> db.createCollection("emp")
    db.emp.insertOne({eno:1,ename:"johnDoe",
salary:60000,desig:"software",
dept:{deptno:10,deptname:"IT",
location:"new
york"}},project:
    {pname:"pp1",hrs:5}})

db.emp.find({salary:{$gt:50000,$lt:80000}})

db.emp.updateMany({"project.pname":"pp1"},{$
set:{"project.hrs":7}})

```

NOTES

- Bootstrap
bootstrap is a free open source frontend framework
provides prebuilt html css js component for building mobile websites
- Jumbotron in Bootstrap
jumbtron- is a bootstrap component that highlights key content or info

- bootstrap advanced component
navigation bars - placed at the top of the webpage

use for display

menu, links, forms

- alerts - to provide user feedback
display user friendly

msgs

modals - a popup window commonly used for login forms, sign-up

or display additional info

- button group - joins multiple elements

- cards - images, headers, footers

- Accordion - used for FAQ or detailed

info

- carousel - display slideshow of

imgs or content

- dropdown - creates a menu of

actions, links, or forms

- list groups - displaying a series of component

jumbotron- is a bootstrap component that highlights key content or info

- benefits of using Bootstrap

faster development - bootstrap provides a set of prebuilt

components(buttons, forms, navigation bars)

responsive and mobile first design - the

framework is built with a mobile
first approach
flexible grid system that automatically
adjusts

layouts to fit
various screen sizes
consistency - maintain look. standardized
styles and components

ease of use - the developer with basic
knowledge of html and css

quickly learn and start
using bootstrap

extensive documentation and community
supports - the framework has well
maintained official

documentation

customization - provide default
styles, bootstrap is highly customizable

browser compatibility - websites built
with bootstrap are compatible

with all web browsers
firefox, chrome, google, safari, edge

- 2D and 3D transformation in CSS3

2D - 2D transforms elements into 2
dimensional plane using x(horizontal) and y
(vertical) axes

translate(x,y): moves an element from its
original position to x and y distance

rotate(angle): rotate an element clockwise

around its origin by a given angle

scale(x,y):increase or decrease the size of an element with width x or height y

skew(ax,ay)

matrix():combines all 2D transform functions into a single function

3D - 3D transforms elements into 2 dimensional plane

rotateX(angle):rotates an element around its horizontal x axis

rotateY(angle):rotates an element around its vertical Y axis

rotateZ(angle):rotates an element around its Z axis which is perpendicular to screen

translate(z) :moves an element along the Z axis

scale3d(x,y,z):scales an element in all 3 dimensions

transform_style

backface-visibilityhides the back face of an element

- Master-Slave Replication

is a technique where data is replicated from a primary node(master)

to one or more secondary nodes(slaves)

- column family database

stores data in cols in the form of rows ,optimized for querying large datasets

- Document Database Use cases

content management systems -
articles,blogs,posts,user comments,media
e-commerce product catalogs - book,tv
screen

user profiles and personalization - social
networks,gaming manage user profiles

internet of things and sensor data -
mobile and web application

- Peer to Peer Replication

allows multiple nodes to accept writes and
replcate data with each other

- NOSQL databases.

NoSQL databases offer flexible schema
deigns and scalability

- Cypher - is the quey language ued in neo4j
to query and manipulate graph data

- Replica - a replica is a copy of a
database or a subset of its data

stored on a single node

distributed database system

the process of creating and
maintaining this copies is called
replication

- feature of Distributed Database.

scalability

high availability and fault tolerance

performance and reduced latency

data transparency

distributed query and transaction

processing

modularity and flexibility

replication

sharding

- Class

a class is a blueprint or template for creating objects

it defines the structure

- ID

an id(identifier) is a unique value assigned to each record,row or object within a database

- Collection

is a group of mongodb document

it is an equivalent to rdbms table

a collection exists with in a single db